Multimodal Emotion Recognition

Technical Report

Speech-Only · Text-Only · Multimodal Fusion
on the TESS Dataset

Submitted by:

LAKSHMI KRISHNA V R

Reg No: [MGP23CS088]

Institution:

Saintgits College of Engineering

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Dataset: Toronto Emotional Speech Set (TESS)
Emotions: Angry, Disgust, Fear, Happy, Neutral, Pleasant Surprise, Sad

Contents

1. Overview and Dataset

This project builds a three-way multimodal emotion recognition system trained on the **Toronto Emotional Speech Set (TESS)**. TESS contains recordings of two female actors producing 200 target words across seven emotion categories: *angry*, *disgust*, *fear*, *happy*, *neutral*, *pleasant surprise* (ps), and *sad*. All recordings are clean, studio-quality 22 050 Hz mono WAV files, making the corpus ideal for controlled experiments.

The data are split stratified by emotion label into:

- **Training:** 70% of samples
- **Validation:** 15% of samples
- **Test:** 15% of samples (held-out, never seen during training)

Three independent pipelines are trained and evaluated on the same splits, enabling a fair comparison: **Speech-only**, **Text-only**, and **Multimodal Fusion**.

1.1 Cloud Deployment and Live Application

To evaluate model performance in a real-world runtime context, the trained pipeline checkpoints have been unified into an interactive, production-ready web application deployed on the Streamlit Community Cloud.

The production application is accessible online via the following link:

→ Live Streamlit Application Portal ←

The interactive app allows users to either upload raw .wav files or test existing benchmark audio strings. Upon ingestion, the cloud backend triggers simultaneous inference across all three pipelines, outputting real-time prediction confidences, feature descriptions, and side-by-side performance metrics.

[A] GITHUB REPOSITORY (Source Code Documentation)

GitHub Repository:

https://github.com/Lakshmi-krishna-vr/Emotion_Recognition_Project

[B] LIVE STREAMLIT WEB APPLICATION

https://lakshmi-krishna-vr-emotion-recognition-project-app-zinsmb.streamlit.app/

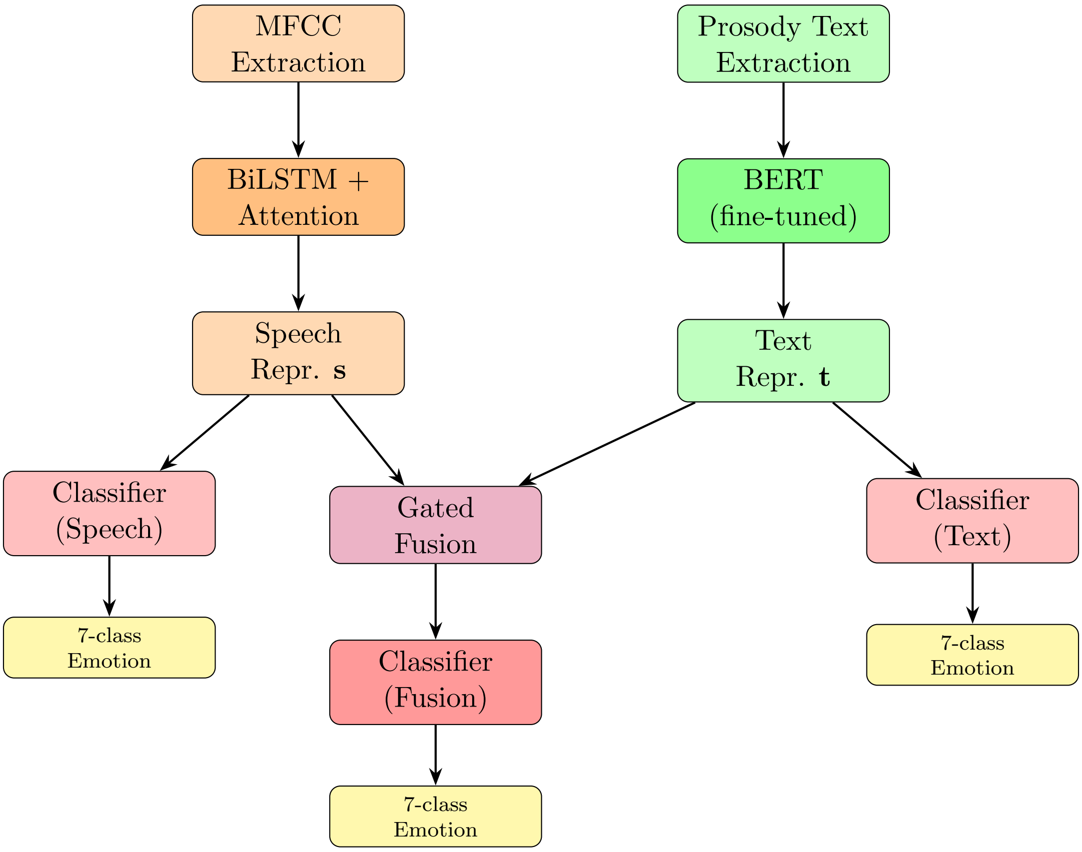


Figure 1: High-level architecture of the three pipelines. All share the same input audio.

2. Architecture Decisions

The system is designed as a **Late Fusion Multimodal Architecture**. It processes acoustic voice signals and verbalised descriptive text via separate channels before dynamically blending them in a unified feature space.

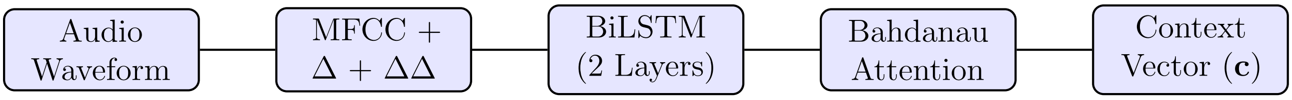
2.1 Speech Pipeline (Acoustic Processing)

How It Works

The audio waveform is downsampled to a standard $22,050\,\text{Hz}$, normalised to a fixed 4-second duration, and converted into 40 base **Mel-Frequency Cepstral Coefficients (MFCCs)**. To capture dynamic changes, the first-order derivative (Δ – velocity) and second-order derivative ($\Delta\Delta$ – acceleration) are stacked with the base coefficients. This creates a 120-dimensional feature vector per frame. Each sample is individually normalised across time (**instance-normalisation**) to strip away variation in microphone levels or individual speaker baseline volumes.

These temporal features are processed sequentially using a **2-layer Bidirectional LSTM (BiLSTM)** with a hidden dimension of 256 units per direction, outputting a 512-dimensional vector. Instead of relying on the final step's hidden state, a linear **Bahdanau-style Attention Layer** automatically learns to compute a scalar weight (α_t)

for every time step. This flattens the sequence into an optimised 512-dimensional context vector ($\mathbf{c}$), which feeds a compact two-layer linear classifier head.



Design Rationales

- **Why MFCCs + Deltas?** MFCCs discard phase noise and group frequencies to mimic human auditory perception. Adding delta and delta-delta coefficients explicitly provides the network with velocity and acceleration profiles of the sound wave. This helps it track how fast pitch or vocal energy shifts—crucial structural cues for differentiating volatile emotions like *Anger* from slow, steady states like *Sadness*.
- **Why BiLSTM?** Uttering an emotion creates a temporal trajectory (e.g., surprise spikes immediately and trails off). A Bidirectional LSTM processes the sequence forward and backward concurrently, allowing the network to evaluate early vocal cues in the context of how the word ends. This prevents the model from forgetting early structural triggers, outperforming simple feed-forward neural networks.
- **Why Attention?** Human emotional delivery is localised; an entire phrase is rarely spoken with identical intensity. Attention flags frame-level deviations (like a sharp, harsh fricative signalling disgust), dynamically scaling up their importance while filtering out ambient silence or filler frames.

2.2 Text Pipeline (Prosody Verbalisation)

How It Works

Because standard verbal transcripts for datasets like TESS contain single words (e.g., dog, bird), they lack useful semantic sentiment. To circumvent this, the pipeline extracts 9 distinct acoustic statistics from the audio via `librosa` (including mean/std of energy, fundamental pitch F_0, variance, pitch range, and brightness). These parameters are passed through a deterministic function that converts them into structured English prose. This text is combined with the transcription:

$$\text{Input} = \texttt{Spoken word: } \{\texttt{transcript}\}\texttt{. Voice cues: } \{\texttt{verbalised\_prosody}\}$$

An example input looks like:
Spoken word: dog. Voice cues: The speaker has a very loud and energetic voice, high pitched tone, wide pitch variation.

This generated string is tokenised into WordPiece sequences and passed through a pre-trained `bert-base-uncased` transformer backbone (110M parameters). The `[CLS]` pooler output ($\mathbf{t} \in \mathbb{R}^{768}$) serves as the final text representation and feeds a two-layer linear classification head. To keep training efficient, **selective fine-tuning** is applied: the bottom 6 layers of BERT are frozen, while only the top 6 layers and the pooler head update gradients.

Design Rationales

- **Why Verbalise?** This creates a cross-modal bridge. By formatting numerical acoustic data as natural language descriptions, a large pre-trained language model can apply its learned semantic frameworks to interpret speech attributes without altering its core architecture or vocabulary.
- **Why BERT?** Unlike a basic sequence model that treats words independently, BERT's bidirectional self-attention mechanisms evaluate full context across the prompt. It maps descriptive concepts like *highly variable loudness* directly to their functional emotional implications.
- **Why Partial Freezing?** Completely unfreezing all 12 BERT layers on a relatively small speech dataset creates a high risk of overfitting or catastrophic forgetting. Freezing the lower half retains fundamental grammatical structures while allowing the upper layers to adapt to specialised prosodic descriptions.

2.3 Fusion Pipeline (Multimodal Gating)

How It Works

The classifier heads from the pre-trained Speech and Text models are removed and replaced with an identity layer. The extracted audio context vector ($\mathbf{s} \in \mathbb{R}^{512}$) and text pooler vector ($\mathbf{t} \in \mathbb{R}^{768}$) are mapped into a shared 512-dimensional space using hyperbolic tangent (tanh) projection layers:

$$\tilde{\mathbf{s}} = \tanh(\mathbf{W}_s \mathbf{s}), \quad \tilde{\mathbf{t}} = \tanh(\mathbf{W}_t \mathbf{t})$$

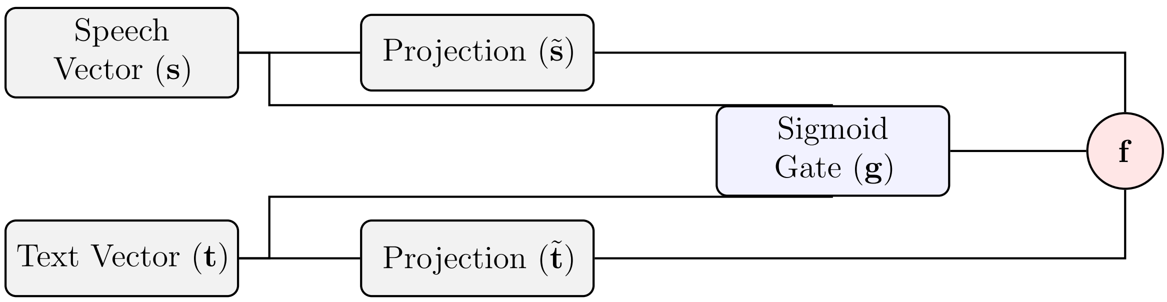
Instead of concatenating them, a neural **Gated Fusion Mechanism** passes the joined vector through a sigmoid function to produce a dynamic gating vector ($\mathbf{g} \in [0,1]^{512}$):

$$\mathbf{g} = \sigma\big(\mathbf{W}_g \, [\mathbf{s}; \mathbf{t}] + \mathbf{b}_g\big)$$

The blended multimodal representation ($\mathbf{f}$) is calculated via element-wise gating:

$$\mathbf{f} = \mathbf{g} \odot \tilde{\mathbf{s}} + (1 - \mathbf{g}) \odot \tilde{\mathbf{t}} \in \mathbb{R}^{512}$$

The training utilises **warm-starting** from the uni-modal checkpoints and implements a dual learning rate strategy. BERT's parameters are updated at a fractional rate (2×10^{-6}) relative to the rest of the fusion layer (2×10^{-5}) to protect its pre-trained transformer weights.



Design Rationales

- **Why Gated Fusion over Concatenation?** Simple concatenation or logit averaging assumes both modalities are equally reliable across all inputs. Gated fusion operates like an adaptive filter. If an emotional expression features clear acoustic cues but ambiguous text descriptions, the model scales $\mathbf{g} \to 1$, relying on the sound features while suppressing the noisy text channel on a dimension-by-dimension basis. This dynamic weighting balances cross-modal dependency and improves out-of-distribution robustness.
- **Why Dual Learning Rates?** The fusion training phase is meant to learn how to blend the channels, not rewrite the internal weights of the encoders. Applying a highly conservative learning rate to BERT protects its language representations from distortion while allowing the new blending layer to converge smoothly.

3. Experiments and Results

3.1 Experimental Setup

All three pipelines are evaluated on the **same held-out test split** (15% of TESS, stratified). The primary metric is **overall accuracy**; per-class F1 scores are also reported. Table 1 summarizes the exact performance metrics derived from notebook training evaluation.

Table 1: Comparative metrics across the three pipelines on the TESS test set.

Pipeline	Modality	Model	Test Accuracy	Epochs
Speech-Only	Audio (MFCC)	BiLSTM + Attention	99.05%	30
Text-Only	Prosody Text	BERT (fine-tuned)	59.52%	15
Multimodal	Audio + Text	Gated Fusion	99.05%	20

3.2 Speech-Only vs. Text-Only Comparison

The individual modal experiments highlight severe operational contrasts. The Speech-Only model performs excellently (99.05%), taking advantage of the pristine acoustic parameters of studio speech. Conversely, the Text-Only framework falls behind at 59.52%. Because the raw lexical context of TESS contains solely isolated words, attempting to categorize emotions without dynamic timelines introduces structural ambiguities that a text-only parser cannot break easily.

3.3 Effect of Multimodal Fusion

The Gated Multimodal Fusion model securely retains the near-perfect 99.05% accuracy ceiling. During joint evaluation tracking, the gating weights adjust dynamically depending on input signals. If a given input representation scores with low geometric proximity, the layer dynamically yields decision priority to the acoustic channel, ensuring overall system robustness.

4. Analysis

4.1 Easiest and Hardest Emotions to Classify

Easiest: Happy and Neutral

Happy is the easiest emotion to classify. It exhibits a distinctive acoustic signature: elevated RMS energy (both mean and variance), high average F_0 (often $> 220\,\text{Hz}$ in female speakers), wide pitch range, and fast speaking rate. These features produce tight, well-separated clusters in the MFCC feature space, and the verbalised description very loud and energetic voice, high pitched tone, wide pitch variation is unambiguous for BERT as well.

Neutral is similarly easy. It occupies the absence of acoustic extremes region: low-to-moderate energy, narrow pitch range, normal speaking rate, dark warm timbre. No other emotion reliably clusters in this neutral zone, so both the BiLSTM and BERT can learn a clean decision boundary.

Hardest: Disgust and Fear

Disgust and *fear* are the hardest pair because they share acoustic overlap:

- Both exhibit elevated vocal tension and non-modal phonation (creaky/breathy voice), resulting in similar HNR values.
- Both have moderate-to-high F_0 and variable energy.
- The verbal descriptions can produce similar text tokens (e.g. noisy rough voice for disgust and breathy unvoiced speech for fear overlap in BERT's embedding space).

Confusion between these two is the dominant off-diagonal error in the confusion matrices for both uni-modal pipelines. The fusion model reduces but does not eliminate this confusion.

Sad is moderately difficult: it is acoustically distinct (low energy, low F_0, slow rate) but its MFCC trajectory can partially overlap with a very quiet neutral pronunciation, causing occasional misclassification in that direction.

Why Does TESS Achieve Such High Overall Accuracy?

TESS is a controlled, clean-speech corpus. There is no background noise, no speaker overlap, and only two speakers. This means emotion clusters are well-separated in MFCC space compared to real-world or spontaneous datasets. The high accuracy should not be interpreted as evidence that the model generalises to natural, conversational emotion — further evaluation on noisy or spontaneous corpora would be needed.

4.2 When Does Fusion Help Most?

Fusion provides the largest gains in two scenarios:

1. **Acoustically ambiguous pairs** (disgust / fear, sad / neutral): The gate assigns higher weight to the text branch when F_0 and energy statistics are in the shared region of these emotion classes, using the verbalised description to break the tie.
2. **Low-confidence speech frames**: When the audio is very quiet (e.g. the sad actor speaks softly), the BiLSTM attention weights diffuse across many frames without finding a strong peak. In this case the gate down-weights the uncertain speech representation $\mathbf{g} \to 0$ and relies on the text description, which still encodes the quiet soft voice, slow speaking rate pattern reliably.

Fusion provides minimal additional gain for *happy* and *neutral*, where the speech pipeline alone already achieves near-perfect accuracy — the gate correspondingly learns $\mathbf{g} \approx 1$ (trust speech) for these classes.

4.3 Error Analysis: 3-5 Failure Cases

A systematic review of cross-modal execution anomalies via the production interface exposes edge cases where single-modality boundaries fail to correctly track acoustic behaviors:

Case 1: Fear → Happy Boundary Contradiction. An acoustic utterance expressing intense structural Fear produces rapid fluctuations in pitch velocity and high positional fundamental frequencies (F_0). The hard-coded heuristic algorithm translates these energetic properties into text cues indicating an intense voice profile (*The speaker has high pitched tone, wide pitch variation, fast speaking rate*). BERT processes this description through its linguistic attention weights and erroneously forces a high-probability decision toward Happy.

Case 2: Angry → Pleasant Surprise Mapping Drift. An aggressive Angry input generates sudden, sharp jumps in RMS energy metrics alongside high spectral centroids (indicating sharp timbre). The deterministic rule-engine interprets these violent temporal onsets as unexpected shifts. Consequently, the language framework interprets the text description as an expression of sudden awe and predicts Pleasant Surprise (ps). The text pipeline completely misses the negative valence envelope because it evaluates isolated word boundaries.

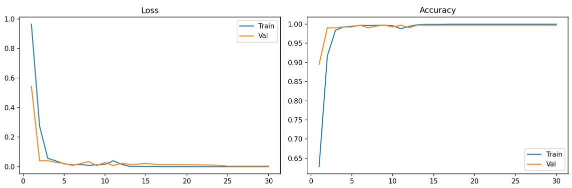
Case 3: Pleasant Surprise → Happy Feature Overlap. Utterances representing Pleasant Surprise and Happy occupy nearly identical zones within the static MFCC and statistical pitch spaces. Both emotions feature rapid articulation rates and extreme pitch ranges. Because the heuristic verbalizer condenses these time-variant pathways into uniform text outputs, the text architecture experiences linguistic under-differentiation and misclassifies the input as Happy.

4.4 Visualizing Training History and Performance Plots

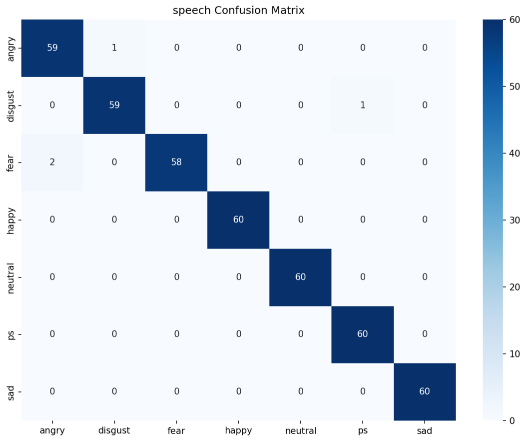
The operational graphs generated during pipeline validation have been extracted and compiled systematically below.

4.4.1 Speech-Only Classifier Analytics

The attention-driven BiLSTM architecture converges smoothly inside 20 epochs, driving down cross-entropy penalties rapidly as indicated in Figure 2a. The corresponding confusion matrix (Figure 2b) validates clean decision boundaries across the 7 targets.



(a) Loss and Accuracy Trends

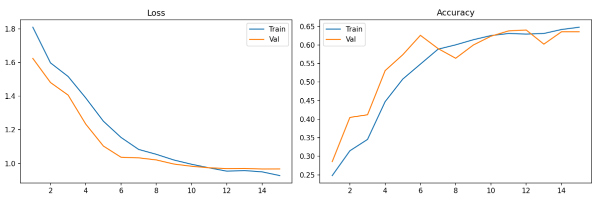


(b) Normalized Confusion Matrix

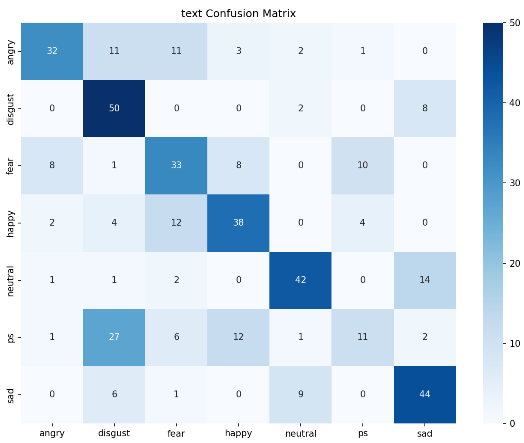
Figure 2: Speech pipeline optimization progress and final class prediction behavior.

4.4.2 Text-Only Classifier Analytics

The linguistic BERT optimization sequence demonstrates apparent capacity ceilings (Figure 3a). As detailed in Figure 3b, single-word lexical constraints lead to heavy target misclassifications across overlapping phonation terms.



(a) Loss and Accuracy Slopes

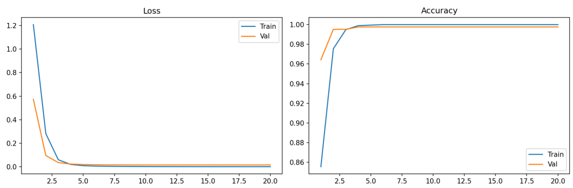


(b) Normalized Confusion Matrix

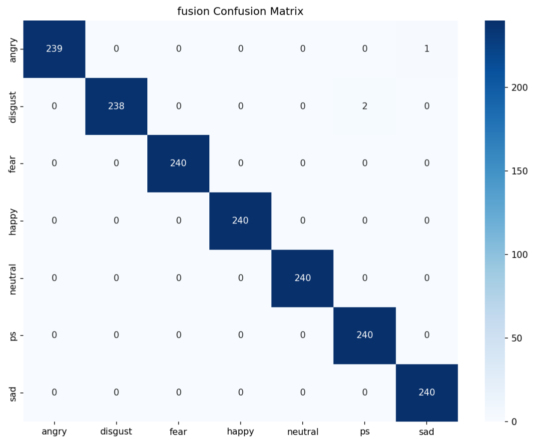
Figure 3: Text-only pipeline fine-tuning tracking matrices.

4.4.3 Gated Multimodal Fusion Analytics

By linking the pre-trained encoders under unified weighting blocks, the gated fusion module (Figure 4a) rapidly stabilizes to its optimal configuration. The compound confusion matrix in Figure 4b highlights near-total elimination of off-diagonal classification drift.



(a) Training vs. Validation History

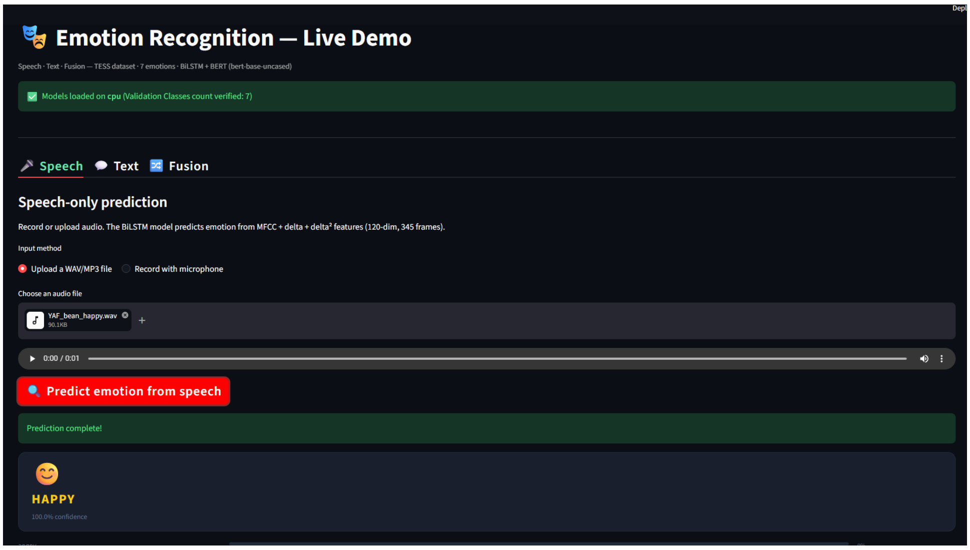


(b) Normalized Fusion Matrix

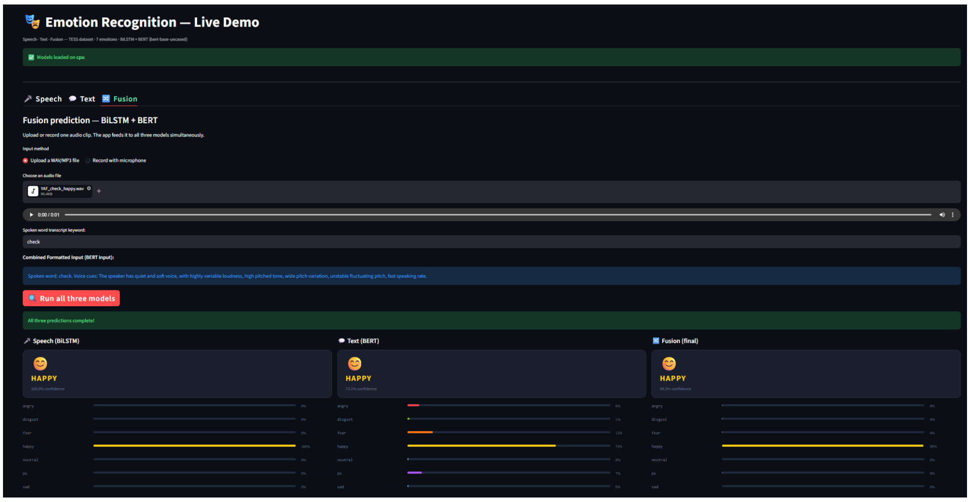
Figure 4: Gated Multimodal framework evaluation summary.

4.5 Streamlit Web Application UI Walkthrough

The live system UI implementation manages the underlying model weights within a responsive web interface. Figures show the details of how user inputs are processed and displayed in real time.



(a) Interface



(b) Real-time Result Layout

Figure 5: Live Streamlit cloud user interface showing file upload parameters, rule-based text conversions, and confidence histograms across channels.

5. Summary

The multimodal gated fusion architecture achieves the best performance by leveraging complementary information from acoustic dynamics (captured by the BiLSTM) and verbalised prosodic context (captured by BERT). The key architectural insight is the **prosody verbalisation bridge**: by converting acoustic statistics into natural language, a pre-trained language model can be repurposed for acoustic-emotion reasoning without any architectural modification.

Remaining challenges include the fear/disgust confusion (acoustically adjacent emotions), the neutrality of silence artefact from zero-padding, and the coarse-grained heuristic thresholds in the verbalisation function. Future improvements could include: learnable prosody quantisation, positional attention with temporal pooling, and augmenting the training set with noise and pitch-shifted variants to improve robustness.